MEENATCHI SUNDARAM K

AI ML Engineer | Generative AI | Agentic AI | LLM | RAG | NLP | Python

Chennai, Tamil Nadu • 8667856836 • enigmasun326@gmail.com • linkedin.com/in/meenatchi-sundaram953457188 • github.com/Meenatchi7502

PROFESSIONAL SUMMARY

AI/ML Engineer with around 3 years of experience designing, developing, and deploying intelligent systems, LLM-based solutions, and Generative AI applications on large-scale datasets. Proficient in Python, LangChain, LangGraph, RAG pipelines, Agentic AI, Prompt Engineering, NLP, and vector databases. Hands-on experience building end-to-end ML pipelines, automating data workflows, and deploying AI-driven solutions into real-world business applications. Proven track record of delivering measurable impact — improving model efficiency by 35%, reducing processing errors by 40%, and enabling data-driven decision-making at scale.

CORE COMPETENCIES

Large Language Models (LLMs) | Generative AI | Agentic AI | NLP | Retrieval-Augmented Generation (RAG) | Prompt Engineering | Semantic Search | AI Chatbots | Conversational AI | Text Classification | Named Entity Recognition (NER) | LangGraph | Multi-Agent Orchestration | Feature Engineering | ML Pipelines | Vector Databases | REST APIs | MLOps | Document Intelligence

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, SpaCy, NLTK), SQL, PySpark (basics)

Generative AI: LLMs, GPT-4, LLaMA, RAG, Embeddings, Prompt Engineering, LangChain, LangGraph, LlamaIndex, Agentic AI, LoRA/QLoRA Fine-Tuning

AI / ML: NLP, Text Classification, NER, Feature Engineering, XGBoost, Random Forest, Model Training & Evaluation, MLOps, Deep Learning

Vector Databases: FAISS, Pinecone, ChromaDB

Frameworks: HuggingFace Transformers, SentenceTransformers, FastAPI, Streamlit, TensorFlow, PyTorch, Keras

Tools & Platforms: Git, Docker, MLflow, Jupyter Notebook, Apache Airflow, CI/CD Pipelines, AWS

Databases: PostgreSQL, MySQL

PROFESSIONAL EXPERIENCE

AI ML Engineer | NielsenIQ — Chennai, India *Jul 2024 – May 2026*

- Built and maintained machine learning models on large-scale FMCG client datasets using Python and Scikit-learn, supporting downstream business reporting and analytics workflows.
- Built RAG pipelines using LangChain integrated with FAISS and ChromaDB vector databases to enable semantic document retrieval and automated business query resolution.
- Designed multi-step Agentic AI workflows using LangChain and LangGraph, enabling task decomposition, tool-use orchestration, and context-aware reasoning across business use cases.
- Implemented LLM-based solutions using GPT-4 and prompt engineering techniques (few-shot, chain-of-thought) to automate structured data extraction from unstructured documents.
- Applied NLP techniques, including text classification and Named Entity Recognition, to process unstructured business datasets and support downstream decision-making pipelines.
- Exposed AI workflows as REST APIs using FastAPI, enabling integration of RAG and LLM-based automation into existing internal business tools.
- Evaluated and iterated on prompt templates and retrieval configurations by comparing outputs against sample query sets, working with cross-functional data and engineering teams to refine automation reliability.

Management Trainee | Niwas Housing Finance — Tenkasi, India *Jun 2023 – Feb 2024*

- Analyzed financial and customer datasets for loan applications on a monthly basis to identify data quality issues and support operational decision-making.
- Built automated reporting workflows using Python and Excel-based tools to generate recurring customer and portfolio insights.

- Managed workflow tracking and documentation to support compliance with internal operational standards.

KEY ACHIEVEMENTS

- Improved ML model inference efficiency on large-scale FMCG client datasets through optimised pipeline design.
- Reduced workflow reporting errors by integrating LangChain-powered AI automation into cross-functional processes.
- Built production-ready RAG pipeline using LangChain, LangGraph, and FAISS enabling real-time document intelligence.
- Achieved strong NLP classification accuracy on document intelligence workflows using chain-of-thought prompt engineering.
- Reduced financial dataset processing errors through rigorous data validation and ML-driven automation.

PROJECTS

1. LLM-Powered FAQ & Document Intelligence Assistant

Stack: Python, LangChain, LangGraph, OpenAI GPT-4, FAISS, HuggingFace, FastAPI

- Designed and built a RAG pipeline using LangChain and OpenAI embeddings to answer domain-specific queries from a large document corpus with high accuracy.
- Implemented Agentic AI workflows using LangGraph for multi-step autonomous query resolution and context-aware response generation.
- Applied chain-of-thought prompt engineering and semantic search with FAISS to improve answer relevance and retrieval accuracy.
- Deployed the assistant via FastAPI with a REST endpoint for real-time query handling and integrated a Streamlit-based human-in-the-loop review interface.

2. AI-Powered Document Validation & Classification System

Stack: Python, GPT-4, LangChain, ChromaDB, Prompt Engineering, Streamlit, Scikit-learn

- Automated document processing and classification using LangChain with chain-of-thought prompt engineering, achieving strong classification accuracy across multiple document categories.
- Built intelligent document Q&A and summarisation pipelines using LlamaIndex and ChromaDB vector database for semantic retrieval.
- Integrated XGBoost and Random Forest models for structured data validation alongside LLM-based unstructured document analysis.
- Deployed a Streamlit-based interface with confidence scoring, reducing manual review turnaround time.

3. Fraud Detection & Anomaly Intelligence Engine

Stack: Python, PyTorch, Scikit-learn, XGBoost, Isolation Forest, LLMs, PostgreSQL

- Built a multi-layer fraud detection system using Scikit-learn Isolation Forest and XGBoost to flag anomalous patterns in real time with low latency.
- Integrated LLM-based summarisation to auto-generate anomaly reports from transaction signals, reducing manual review effort.
- Applied NLP-based NER and text classification to extract entities from unstructured financial records, improving detection accuracy.
- Deployed the pipeline on AWS with Docker containerisation, achieving high uptime and a meaningful reduction in false positives.

EDUCATION

Master of Business Administration (MBA) | Bharathiar University 2021 – 2023
Bachelor of Computer Application (BCA) | RVS College of Arts and Science 2017 – 2020